

Runtime changes for migration to .NET Framework 4.5.x

Runtime changes for migration to .NET Framework 4.5.x - chlowell, learn.microsoft.com.

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Content

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Runtime changes for migration to .NET Framework 4.5.x

Summarize this article for me

This article lists the app compatibility issues that were introduced in .NET Framework [4.5](), [4.5.1](), and [4.5.2]().

.NET Framework 4.5

ASP.NET

GridViews with AllowCustomPaging set to true may fire the PageIndexChanging event when leaving the final page of the view

Details

A bug in the .NET Framework 4.5 causes [System.Web.UI.WebControls.GridView.PageIndexChanging](#) to sometimes not fire for [System.Web.UI.WebControls.GridViews](#) that have enabled [System.Web.UI.WebControls.GridView.AllowCustomPaging](#).

Suggestion

This issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework. As a work-around, the app can do an explicit `BindGrid` on any `Page_Load` that would hit these conditions (the `System.Web.UI.WebControls.GridView` is on the last page and `LastSystem.Web.UI.WebControls.GridView.PageSize` is different from `System.Web.UI.WebControls.GridView.PageSize`). Alternatively, the app can be modified to allow paging (instead of custom paging), as that scenario does not demonstrate the problem.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- [GridView.AllowCustomPaging](#)

HttpRequest.ContentEncoding property prohibits UTF7

Details

Beginning in .NET Framework 4.5, UTF-7 encoding is prohibited in `System.Web.HttpRequests`' bodies. Data for applications that depend on incoming UTF-7 data will not decode properly in some cases.

Suggestion

Ideally, applications should be updated to not use UTF-7 encoding in `System.Web.HttpRequests`. Alternatively, legacy behavior can be restored by using the `aspnet:AllowUtf7RequestContentEncoding` attribute of the `appSettings` element.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

- [HttpRequest.ContentEncoding](#)

HttpUtility.JavaScriptStringEncode escapes ampersand

Details

Starting with the .NET Framework 4.5, `System.Web.HttpUtility.JavaScriptStringEncode(String)` escapes the ampersand (&) character.

Suggestion

If your app depends on the previous behavior of this method, you can add an `aspnet:JavaScriptDoNotEncodeAmpersand` setting to the `ASP.NET appSettings element` in your configuration file.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- [HttpUtility.JavaScriptStringEncode\(String\)](#)

- [HttpUtility.JavaScriptStringEncode\(String, Boolean\)\)](#)

IPad should not be used in custom capabilities file because it is now a browser capability

Details

Beginning in .NET Framework 4.5, iPad is an identifier in the default ASP.NET browser capabilities file, so it should not be used in a custom capabilities file

Suggestion

If iPad-specific capabilities are required, it is necessary to modify iPad behavior by setting capabilities on the pre-defined gateway refID "IPad" instead of by generating a new "IPad" ID by user agent matching.

Name	Value		Scope	Edge		Version	4.5		Type	Runtime	
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Page.LoadComplete event no longer causes System.Web.UI.WebControls.EntityDataSource control to invoke data binding

Details

The [LoadComplete](#) event no longer causes the [System.Web.UI.WebControls.EntityDataSource](#) control to invoke data binding for changes to create/update/delete parameters. This change eliminates an extraneous trip to the database, prevents the values of controls from being reset, and produces behavior that is consistent with other data controls, such as [System.Web.UI.WebControls.SqlDataSource](#) and [System.Web.UI.WebControls.ObjectDataSource](#). This change produces different behavior in the unlikely event that applications rely on invoking data binding in the [LoadComplete](#) event.

Suggestion

If there is a need for databinding, manually invoke databind in an event that is earlier in the post-back.

Name	Value		Scope	Edge		Version	4.5		Type	Runtime	
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Affected APIs

Not detectable via API analysis.

Profiling ASP.NET MVC4 apps can lead to Fatal Execution Engine Error

Details

Profilers using NGEN /Profile assemblies may crash profiled ASP.NET MVC4 applications on startup with a 'Fatal Execution Engine Exception'

Suggestion

This issue is fixed in the .NET Framework 4.5.2. Alternatively, the profiler may avoid this issue by specifying `COR_PRF_DISABLE_ALL_NGEN_IMAGES` in its event mask.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

Not detectable via API analysis.

Sharing session state with ASP.NET StateServer requires all servers in the web farm to use the same .NET Framework version

Details

When enabling [System.Web.SessionState.SessionStateMode.StateServer](#) session state, all of the servers in the given web farm must use the same version of the .NET Framework in order for state to be properly shared.

Suggestion

Be sure to upgrade .NET Framework versions on web servers that share state at the same time.

Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

- [SessionStateMode.StateServer](#)

WebUtility.HtmlDecode no longer decodes invalid input sequences

Details

By default, decoding methods no longer decode an invalid input sequence into an invalid UTF-16 string. Instead, they return the original input.

Suggestion

The change in decoder output should matter only if you store binary data instead of UTF-16 data in strings. To explicitly control this behavior, set the

`aspnet:AllowRelaxedUnicodeDecoding` attribute of the [appSettings](#) element to `true` to enable legacy behavior or to `false` to enable the current behavior.

Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- [WebUtility.HtmlDecode\(String\)](#)
- [WebUtility.HtmlDecode\(String, TextWriter\)](#)
- [WebUtility.UrlDecode\(String\)](#)

Core

Assemblies compiled with `Regex.CompileToAssembly` breaks between 4.0 and 4.5

Details

If an assembly of compiled regular expressions is built with the .NET Framework 4.5 but targets the .NET Framework 4, attempting to use one of the regular expressions in that assembly on a system with .NET Framework 4 installed throws an exception.

Suggestion

To work around this problem, you can do either of the following:

- Build the assembly that contains the regular expressions with the .NET Framework 4.
- Use an interpreted regular expression.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- `[Regex.CompileToAssembly(RegexCompilationInfo[], AssemblyName)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly\(system-text-regularexpressions-regexcompilationinfo\(\)-system-reflection-assemblyname\)](https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly(system-text-regularexpressions-regexcompilationinfo()-system-reflection-assemblyname)))
- `[Regex.CompileToAssembly(RegexCompilationInfo[], AssemblyName, CustomAttributeBuilder[])]`([https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly\(system-text-regularexpressions-regexcompilationinfo\(\)-system-reflection-assemblyname-system-reflection-emit-customattributebuilder\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly(system-text-regularexpressions-regexcompilationinfo()-system-reflection-assemblyname-system-reflection-emit-customattributebuilder())))
- `[Regex.CompileToAssembly(RegexCompilationInfo[], AssemblyName, CustomAttributeBuilder[], String)]`([https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly\(system-text-regularexpressions-regexcompilationinfo\(\)-system-reflection-assemblyname-system-reflection-emit-customattributebuilder\(\)-system-string\)](https://learn.microsoft.com/en-us/dotnet/api/system.text.regularexpressions.regex.compiletoassembly#system-text-regularexpressions-regex-compiletoassembly(system-text-regularexpressions-regexcompilationinfo()-system-reflection-assemblyname-system-reflection-emit-customattributebuilder()-system-string)))

BlockingCollection<T>.TryTakeFromAny does not throw anymore

Details

If one of the input collections is marked completed,

`[TryTakeFromAny(BlockingCollection<T>[], T)]`([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany\(system-collections-concurrent-](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany(system-collections-concurrent-)

`blockingcollection((-0))()-0@))` no longer returns -1 and `[TakeFromAny(BlockingCollection<T>[], T)]` ([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany(system-collections-concurrent-blockingcollection((-0))()-0@)))) no longer throws an exception. This change makes it possible to work with collections when one of the collections is either empty or completed, but the other collection still has items that can be retrieved.

Suggestion

If `TryTakeFromAny` returning -1 or `TakeFromAny` throwing were used for control-flow purposes in cases of a blocking collection being completed, such code should now be changed to use `.Any(b => b.IsCompleted)` to detect that condition.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime	
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Affected APIs

- `[BlockingCollection<T>.TakeFromAny(BlockingCollection<T>[], T)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany(system-collections-concurrent-blockingcollection((-0))()-0@))))
- `[BlockingCollection<T>.TakeFromAny(BlockingCollection<T>[], T, CancellationToken)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@-system-threading-cancellationtoken\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.takefromany#system-collections-concurrent-blockingcollection-1-takefromany(system-collections-concurrent-blockingcollection((-0))()-0@-system-threading-cancellationtoken))))
- `[BlockingCollection<T>.TryTakeFromAny(BlockingCollection<T>[], T)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany(system-collections-concurrent-blockingcollection((-0))()-0@))))
- `[BlockingCollection<T>.TryTakeFromAny(BlockingCollection<T>[], T, Int32)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@-system-int32\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany(system-collections-concurrent-blockingcollection((-0))()-0@-system-int32))))
- `[BlockingCollection<T>.TryTakeFromAny(BlockingCollection<T>[], T, TimeSpan)]`
([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@-system-timespan\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany(system-collections-concurrent-blockingcollection((-0))()-0@-system-timespan))))

- `[BlockingCollection<T>.TryTakeFromAny(BlockingCollection<T>[], T, TimeSpan)]` ([https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany\(system-collections-concurrent-blockingcollection\(\(-0\)\)\(\)-0@-system-timespan\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.collections.concurrent.blockingcollection-1.trytakefromany#system-collections-concurrent-blockingcollection-1-trytakefromany(system-collections-concurrent-blockingcollection((-0))()-0@-system-timespan))))

Change in behavior for Task.WaitAll methods with time-out arguments

Details

`Task.WaitAll` behavior was made more consistent in .NET Framework 4.5. In the .NET Framework 4, these methods behaved inconsistently. When the time-out expired, if one or more tasks were completed or canceled before the method call, the method threw an [System.AggregateException](#) exception. When the time-out expired, if no tasks were completed or canceled before the method call, but one or more tasks entered these states after the method call, the method returned false.

In the .NET Framework 4.5, these method overloads now return false if any tasks are still running when the time-out interval expired, and they throw an [System.AggregateException](#) exception only if an input task was cancelled (regardless of whether it was before or after the method call) and no other tasks are still running.

Suggestion

If an [System.AggregateException](#) was being caught as a means of detecting a task that was cancelled prior to the `WaitAll` call being invoked, that code should instead do the same detection via the `IsCanceled` property (for example: `.Any(t => t.IsCanceled)`) since .NET Framework 4.6 will only throw in that case if all awaited tasks are completed prior to the timeout.

Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- `[Task.WaitAll(Task[], Int32)]` ([https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall\(system-threading-tasks-task\(\)-system-int32\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall(system-threading-tasks-task()-system-int32))))
- `[Task.WaitAll(Task[], Int32, CancellationToken)]` ([https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall\(system-threading-tasks-task\(\)-system-int32-system-threading-cancellationtoken\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall(system-threading-tasks-task()-system-int32-system-threading-cancellationtoken))))
- `[Task.WaitAll(Task[], TimeSpan)]` ([https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall\(system-threading-tasks-task\(\)-system-timespan\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.threading.tasks.task.waitall#system-threading-tasks-task-waitall(system-threading-tasks-task()-system-timespan))))

Compiler support for type forwarding when multi-targeting mscorlib

Details

A new CodeDOM feature allows a compiler to compile against the targeted version of mscorlib.dll instead of the .NET Framework 4.5 version of mscorlib.dll.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

Not detectable via API analysis.

ConcurrentQueue<T>.TryPeek can return an erroneous null via its out parameter

Details

In some multi-threaded scenarios, [System.Collections.Concurrent.ConcurrentQueue<T>.TryPeek\(T\)](#) can return true, but populate the out parameter with a null value (instead of the correct, peeked value).

Suggestion

This issue is fixed in the .NET Framework 4.5.1. Upgrading to that Framework will solve the issue.

Name	Value	Scope	Major	Version	4.5	Type	Runtime
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Affected APIs

- [ConcurrentQueue<T>.TryPeek\(T\)](#)

ETW EventListeners do not capture events from providers with explicit keywords (like the TPL provider)

Details

ETW EventListeners with a blank keyword mask do not properly capture events from providers with explicit keywords. In the .NET Framework 4.5, the TPL provider began providing explicit keywords and triggered this issue. In the .NET Framework 4.6, EventListeners have been updated to no longer have this issue.

Suggestion

To work around this problem, replace calls to [EnableEvents\(EventSource, EventLevel\)](#) with calls to the EnableEvents overload that explicitly specifies the "any keywords" mask to use: `EnableEvents(eventSource, level, unchecked((EventKeywords)0xFFFFffffff))`.

Alternatively, this issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

- [EventListener.EnableEvents\(EventSource, EventLevel\)](#)

Exceptions during unobserved processing in `System.Threading.Tasks.Task` no longer propagate on finalizer thread

Details

Because the [System.Threading.Tasks.Task](#) class represents an asynchronous operation, it catches all non-severe exceptions that occur during asynchronous processing. In the .NET Framework 4.5, if an exception is not observed and your code never waits on the task, the exception will no longer propagate on the finalizer thread and crash the process during garbage collection. This change enhances the reliability of applications that use the Task class to perform unobserved asynchronous processing.

Suggestion

If an app depends on unobserved asynchronous exceptions propagating to the finalizer thread, the previous behavior can be restored by providing an appropriate handler for the [UnobservedTaskException](#) event, or by setting a [runtime configuration element](#).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [Task.Run\(Action\)](#)
- [Task.Run\(Action, CancellationToken\)](#)
- [Task.Run\(Func<Task>\)](#)
- [Task.Run\(Func<Task>, CancellationToken\)](#)-system-threading-cancellationtoken))
- [Task.Run<TResult>\(Func<TResult>\)](#)
- [Task.Run<TResult>\(Func<TResult>, CancellationToken\)](#)-system-threading-cancellationtoken))
- [Task.Run<TResult>\(Func<Task<TResult>>\)](#)
- [Task.Run<TResult>\(Func<Task<TResult>>, CancellationToken\)](#)-system-threading-cancellationtoken))
- [Task.Start\(\)](#)
- [Task.Start\(TaskScheduler\)](#)

List.Sort algorithm changed

Details

Beginning in .NET Framework 4.5, [System.Collections.Generic.List<T>](#)'s sort algorithm has changed (to be an introspective sort instead of a quick sort). [System.Collections.Generic.List<T>](#)'s sort has never been stable, but this change may cause different scenarios to sort in unstable ways. That simply means that equivalent items may sort in different orders in subsequent calls of the API.

Suggestion

Because the old sort algorithm was also unstable (though in slightly different ways), there should be no code that depends on equivalent items always sorting in a particular order. If there are instances of code depending upon that and being lucky with the old behavior, that code should be updated to use a comparer that will deterministically sort the items in the desired order.

Name	Value	Scope	Transparent	Version	4.5	Type	Runtime
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Affected APIs

- [List<T>.Sort\(\)](#)
- [List<T>.Sort\(IComparer<T>\)\)](#)
- [List<T>.Sort\(Comparison<T>\)\)](#)
- [List<T>.Sort\(Int32, Int32, IComparer<T>\)\)](#)

Missing Target Framework Moniker results in 4.0 behavior

Details

Applications without a [System.Runtime.Versioning.TargetFrameworkAttribute](#) applied at the assembly level will automatically run using the semantics (quirks) of the .NET Framework 4.0. To ensure high quality, it is recommended that all binaries be explicitly attributed with a [System.Runtime.Versioning.TargetFrameworkAttribute](#) indicating the version of the .NET Framework they were built with. Note that using a target framework moniker in a project file will cause MSBuild to automatically apply a [System.Runtime.Versioning.TargetFrameworkAttribute](#).

Suggestion

A [System.Runtime.Versioning.TargetFrameworkAttribute](#) should be supplied, either through adding the attribute directly to the assembly or by specifying a target framework in the [project file or through Visual Studio's project properties GUI](#).

Name	Value	Scope	Major	Version	4.5	Type	Runtime
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Affected APIs

Not detectable via API analysis.

Some .NET APIs cause first chance (handled) EntryPointNotFoundExceptions

Details

In the .NET Framework 4.5, a small number of .NET methods began throwing first chance [System.EntryPointNotFoundExceptions](#). These exceptions were handled within the .NET Framework, but could break test automation that did not expect the first chance

exceptions. These same APIs break some ApiVerifier scenarios when HighVersionLie is enabled.

Suggestion

This bug can be avoided by upgrading to .NET Framework 4.5.1. Alternatively, test automation can be updated to not break on first-chance [System.EntryPointNotFoundException](#) exceptions.

		Value			Scope		Edge			Version		4.5			Type		Runtime		
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Affected APIs

- [Debug.Assert\(Boolean\)](#)
- [Debug.Assert\(Boolean, String\)](#)
- [Debug.Assert\(Boolean, String, String\)](#)
- [\[Debug.Assert\(Boolean, String, String, Object\[\]\)\]](#)([https://learn.microsoft.com/en-us/dotnet/api/system.diagnostics.debug.assert#system-diagnostics-debug-assert\(system-boolean-system-string-system-string-system-object\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.diagnostics.debug.assert#system-diagnostics-debug-assert(system-boolean-system-string-system-string-system-object())))
- [XmlSerializer\(Type\)](#)

System.Threading.Tasks.Task no longer throw ObjectDisposedException after object is disposed

Details

Except for [IAsyncResult.AsyncWaitHandle](#), [System.Threading.Tasks.Task](#) methods no longer throw an [System.ObjectDisposedException](#) exception after the object is disposed. This change supports the use of cached tasks. For example, a method can return a cached task to represent an already completed operation instead of allocating a new task. This was impossible in previous .NET Framework versions, because any consumer of the task could dispose of it, which rendered it unusable.

Suggestion

Be aware that Task methods may no longer throw [System.ObjectDisposedException](#) in cases when the object is disposed. If an app was depending on this exception to know that a task was disposed, it should be updated to explicitly check the task's status using [Status](#).

	Name		Value			Scope		Minor			Version		4.5			Type		Runtime		
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Affected APIs

Not detectable via API analysis.

System.Uri escaping now supports RFC 3986

Details

URI escaping has changed in .NET Framework 4.5 to support [RFC 3986](#). Specific changes include:

- [System.Uri.EscapeDataString\(String\)](#) escapes reserved characters based on RFC 3986.
- [System.Uri.EscapeUriString\(String\)](#) does not escape reserved characters.
- [System.Uri.UnescapeDataString\(String\)](#) does not throw an exception if it encounters an invalid escape sequence.
- Unreserved escaped characters are un-escaped.

Suggestion

- Update applications to not rely on [System.Uri.UnescapeDataString\(String\)](#) to throw in the case of an invalid escape sequence. Such sequences must be detected directly now.
- Similarly, expect that Escaped and Unescaped URI and Data strings may vary from .NET Framework 4.0 and .NET Framework 4.5 and should not be compared across .NET versions directly. Instead, they should be parsed and normalized in a single .NET version before any comparisons are made.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
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Affected APIs

- [Uri.EscapeDataString\(String\)](#)
- [Uri.EscapeUriString\(String\)](#)
- [Uri.UnescapeDataString\(String\)](#)

Data

Sql_variant data uses sql_variant collation rather than database collation

Details

`sql_variant` data uses `sql_variant` collation rather than database collation.

Suggestion

This change addresses possible data corruption if the database collation differs from the `sql_variant` collation. Applications that rely on the corrupted data may experience failure.

Name	Value	Scope	Transparent	Version	4.5	Type	Runtime
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Affected APIs

Not detectable via API analysis.

SqlBulkCopy uses destination column encoding for strings

Details

When inserting data into a column, [System.Data.SqlClient.SqlBulkCopy](#) uses the encoding of the destination column rather than the default encoding for `VARCHAR` and `CHAR` types. This change eliminates the possibility of data corruption caused by using the default encoding when the destination column does not use the default encoding. In rare cases, an existing application may throw a `SqlException` exception if the change in encoding produces data that is too big to fit into the destination column.

Suggestion

Expect that [System.Data.SqlClient.SqlBulkCopy](#) will no longer corrupt data due to encoding differences. If strings near the destination column's size limit are being copied, it may be necessary to either pre-encode data (to be copied to check that the data will fit in the destination column) or catch [System.Data.SqlClient.SqlExceptions](#).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

- [System.Data.SqlClient.SqlBulkCopy](#)
- [SqlBulkCopy\(SqlConnection\)](#)

SqlConnection can no longer connect to SQL Server 1997 or databases using the VIA adapter

Details

Connections to SQL Server databases using the [Virtual Interface Adapter \(VIA\) protocol](#) are no longer supported. The protocol used to connect to a SQL Server database is visible in the connection string. A VIA connection will contain `via:<servername>`. If this app is connecting to SQL via a protocol other than VIA (tcp: or np: for example), then no breaking change will be encountered. Also, connections to SQL Server 7 (1997) are no longer supported.

Suggestion

The VIA protocol is deprecated, so an alternative protocol should be used to connect to SQL databases. The most common protocol used is TCP/IP. For more information about connecting through TCP/IP, see [Enable the TCP/IP protocol for a database instance](#). If the database is only accessed from within an intranet, the shared pipes protocol may provide better performance if the network is slow.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
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Affected APIs

- [SqlConnection\(String\)](#)
- [SqlConnection\(String, SqlCredential\)](#)

SqlConnection.Open fails on Windows 7 with non-IFS Winsock BSP or LSP present

Details

[Open\(\)](#) and [OpenAsync\(CancellationToken\)](#) fail in the .NET Framework 4.5 if running on a Windows 7 machine with a non-IFS Winsock BSP or LSP are present on the computer. To determine whether a non-IFS BSP or LSP is installed, use the `netsh WinSock Show Catalog` command, and examine every `Winsock Catalog Provider Entry` item that is returned. If the Service Flags value has the `0x20000` bit set, the provider uses IFS handles and will work correctly. If the `0x20000` bit is clear (not set), it is a non-IFS BSP or LSP.

Suggestion

This bug has been fixed in the .NET Framework 4.5.2, so it can be avoided by upgrading the .NET Framework. Alternatively, it can be avoided by removing any installed non-IFS Winsock LSPs.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime	
------	-------	-------	-------	---------	-----	------	---------	--

Affected APIs

- [SqlConnection.Open\(\)](#)
- [SqlConnection.OpenAsync\(CancellationToken\)](#)

Debugger

Null coalescer values are not visible in debugger until one step later

Details

A bug in the .NET Framework 4.5 causes values set via a null coalescing operation to not be visible in the debugger immediately after the assignment operation is executed when running on the 64-bit version of the Framework.

Suggestion

Stepping one additional time in the debugger will cause the local/field's value to be correctly updated. Also, this issue has been fixed in the .NET Framework 4.6; upgrading to that version of the Framework should solve the issue.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime	
------	-------	-------	------	---------	-----	------	---------	--

Affected APIs

Not detectable via API analysis.

Entity Framework

Change in behavior in Data Definition Language (DDL) APIs

Details

The behavior of DDL APIs when AttachDBFilename is specified has changed as follows:

- Connection strings need not specify an Initial Catalog value. Previously, both AttachDBFilename and Initial Catalog were required.
- If both AttachDBFilename and Initial Catalog are specified and the given MDF file exists, the [DatabaseExists](#) method returns `true`. Previously, it returned `false`.
- If both AttachDBFilename and Initial Catalog are specified and the given MDF file exists, calling the [DeleteDatabase](#) method deletes the files.
- If [DeleteDatabase](#) is called when the connection string specifies an AttachDBFilename value with an MDF that doesn't exist and an Initial Catalog that doesn't exist, the method throws an [InvalidOperationException](#) exception. Previously, it threw a [SqlException](#) exception.

Suggestion

These changes make it easier to build tools and applications that use the DDL APIs. These changes can affect application compatibility in the following scenarios:

- The user writes code that executes a `DROP DATABASE` command directly instead of calling [DeleteDatabase](#) if [DatabaseExists](#) returns `true`. This breaks existing code if the database is not attached but the MDF file exists.
- The user writes code that expects the [DeleteDatabase](#) method to throw a [SqlException](#) rather than an [InvalidOperationException](#) when the Initial Catalog and MDF file don't exist.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Different exception handling forObjectContext.CreateDatabase and DbProviderServices.CreateDatabase methods

Details

Beginning in .NET Framework 4.5, if database creation fails, [CreateDatabase](#) methods will attempt to drop the empty database. If that operation succeeds, the original [System.Data.SqlClient.SqlException](#) will be propagated (instead of the [System.InvalidOperationException](#) that was always thrown in .NET Framework 4.0)

Suggestion

When catching an [System.InvalidOperationException](#) while executing [CreateDatabase\(\)](#) or [CreateDatabase\(DbConnection, Nullable<Int32>, StoreItemCollection\)](#)-system-data-metadata-edm-storeitemcollection)), [SqlExceptions](#) should now also be caught.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [ObjectContext.CreateDatabase\(\)](#)
- [DbProviderServices.CreateDatabase\(DbConnection, Nullable<Int32>, StoreItemCollection\)\)\)-system-data-metadata-edm-storeitemcollection\)\)](#)

EntityFramework 6.0 loads very slowly in apps launched from Visual Studio

Details

Launching an app from Visual Studio 2013 that uses EntityFramework 6.0 can be very slow.

Suggestion

This issue is fixed in EntityFramework 6.0.2. Update EntityFramework to avoid the performance issue.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime	
------	-------	-------	------	---------	-----	------	---------	--

Affected APIs

Not detectable via API analysis.

Log file name created by the ObjectContext.CreateDatabase method has changed to match SQL Server specifications

Details

When the [System.Data.Objects.ObjectContext.CreateDatabase\(\)](#) method is called either directly or by using Code First with the SqlClient provider and an AttachDBFilename value in the connection string, it creates a log file named filename_log.ldf instead of filename.ldf (where filename is the name of the file specified by the AttachDBFilename value). This change improves debugging by providing a log file named according to SQL Server specifications.

Suggestion

If the log file name is important for an app, the app should be updated to expect the standard _log.ldf file name format.

Value	Scope	Edge	Version	4.5	Type	Runtime	
-------	--------------	------	----------------	-----	-------------	---------	--

Affected APIs

- [ObjectContext.CreateDatabase\(\)](#)

ObjectContext.Translate and ObjectContext.ExecuteStoreQuery now support enum type

Details

In .NET Framework 4.0, the generic parameter `T` of `ObjectContext.Translate` and `ObjectContext.ExecuteStoreQuery` methods could not be an enum. That scenario is now supported.

Suggestion

If `Translate` or `ExecuteStoreQuery` was called on an enum type in .NET Framework 4.0, '0' was returned. If that behavior was desirable, the calls should be replaced with a constant 0 (or the enum equivalent of it).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [ObjectContext.Translate<TElement>\(DbDataReader\)](#)
- [ObjectContext.Translate<TEntity>\(DbDataReader, String, MergeOption\)](#)
- [ObjectContext.ExecuteStoreQuery<TElement>\(String, Object\[\]\)](#)
([https://learn.microsoft.com/en-us/dotnet/api/system.data.objects.objectcontext.executestorequery#system-data-objects-objectcontext-executestorequery-1\(system-string-system-object\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.data.objects.objectcontext.executestorequery#system-data-objects-objectcontext-executestorequery-1(system-string-system-object())))
- [ObjectContext.ExecuteStoreQuery<TEntity>\(String, String, MergeOption, Object\[\]\)](#)
([https://learn.microsoft.com/en-us/dotnet/api/system.data.objects.objectcontext.executestorequery#system-data-objects-objectcontext-executestorequery-1\(system-string-system-string-system-data-objects-mergeoption-system-object\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.data.objects.objectcontext.executestorequery#system-data-objects-objectcontext-executestorequery-1(system-string-system-string-system-data-objects-mergeoption-system-object())))

LINQ

Enumerable.Empty<TResult> always returns cached instance

Details

Beginning in .NET Framework 4.5, [Empty<TResult>\(\)](#) always returns a cached internal instance [IEnumerable<T>](#). Previously, [Empty<TResult>\(\)](#) would cache an empty [IEnumerable<T>](#) at the time the API was called, meaning that in some conditions in which [Empty<TResult>\(\)](#) was called rapidly and concurrently, different instances of the type could be returned for different calls to the API.

Suggestion

Because the previous behavior was non-deterministic, code is unlikely to depend on it. However, in the unlikely case that empty enumerables are being compared and expected to sometimes be unequal, explicit empty arrays should be created (`new T[0]`) instead of using [Empty<TResult>\(\)](#).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [Enumerable.Empty<TResult>\(\)](#)

Managed Extensibility Framework (MEF)

MEF catalogs implement IEnumerable and therefore can no longer be used to create a serializer

Details

Starting with the .NET Framework 4.5, MEF catalogs implement IEnumerable and therefore can no longer be used to create a serializer ([System.Xml.Serialization.XmlSerializer](#) object). Trying to serialize a MEF catalog throws an exception.

Suggestion

Can no longer use MEF to create a serializer

Name	Value	Scope	Major	Version	4.5	Type	Runtime	
------	-------	-------	-------	---------	-----	------	---------	--

Affected APIs

Not detectable via API analysis.

Networking

Deserialization of MailMessage objects serialized under the .NET Framework 4.5 may fail

Details

Starting with the .NET Framework 4.5, [MailMessage](#) objects can include non-ASCII characters. In the .NET Framework 4, only ASCII characters are supported. [MailMessage](#) objects that contain non-ASCII characters and that are serialized under the .NET Framework 4.5 or later cannot be deserialized under the .NET Framework 4.

Suggestion

Ensure that your code provides exception handling when deserializing a [MailMessage](#) object.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime	
------	-------	-------	-------	---------	-----	------	---------	--

Affected APIs

- [System.Web.Mail.MailMessage](#)

Details

The System.Net.PeerToPeer.Collaboration namespace is unavailable on Windows 8 or above.

Suggestion

Apps that support Windows 8 or above must be updated to not depend on this namespace or its members.

Name	Value	Scope	Major	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [System.Net.PeerToPeer.Collaboration](#)

Printing

Data written to `PrintSystemJobInfo.JobStream` must be in XPS format

Details

The [JobStream](#) property exposes the stream of a print job. The user can send raw data to the underlying operating system printing components by writing to this stream. Starting with the .NET Framework 4.5 on Windows 8 and later versions of the Windows operating system, data written to this stream must be in XPS format as a package stream.

Suggestion

To output print content, you can do either of the following:

- Use the [XpsDocumentWriter](#) class to output print content. This is the recommended alternative.
- Ensure that the data sent to the stream returned by the [JobStream](#) property is in XPS format as a package stream.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [PrintSystemJobInfo.JobStream](#)

Serialization

BinaryFormatter can fail to find type from LoadFrom context

Details

As of .NET Framework 4.5, a number of [System.Xml.Serialization.XmlSerializer](#) changes may cause differences in deserialization when using [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#) to deserialize types that had been loaded in the LoadFrom context. These changes are due to the new ways [System.Xml.Serialization.XmlSerializer](#) now loads a type which causes different behavior when a [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#) attempts to deserialize to that type later on. The default serialization binder does not automatically search the LoadFrom context, although it may have worked in some circumstances based on the old behavior of XmlSerializer. Due to the

changes, when a type is being loaded from an assembly loaded in a different context, a [System.IO.FileNotFoundException](#) may be thrown.

Warning

Binary serialization with `BinaryFormatter` can be dangerous. For more information, see the [BinaryFormatter security guide](#) and the [BinaryFormatter migration guide](#).

Suggestion

If this exception is seen, the `Binder` property of the [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#) can be set to a custom binder that will find the correct type.

```
var formatter = new BinaryFormatter { Binder = new TypeFinderBinder() }
```

And then the custom binder:

```
public class TypeFinderBinder : SerializationBinder
{
    private static readonly string s_assemblyName =
        Assembly.GetExecutingAssembly().FullName;

    public override Type BindToType(string assemblyName, string typeName)
    {
        return Type.GetType(String.Format(CultureInfo.InvariantCulture, "{0}, {1}",
            typeName, s_assemblyName));
    }
}
```

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#)
- [BinaryFormatter.Deserialize\(Stream\)](#)
- [BinaryFormatter.Deserialize\(Stream, HeaderHandler\)](#)

SoapFormatter cannot deserialize Hashtable and similar ordered collection objects

Details

The [System.Runtime.Serialization.Formatters.Soap.SoapFormatter](#) does not guarantee that objects serialized under one .NET Framework version will successfully deserialize under a different version. Specifically, some ordered collections (like [System.Collections.Hashtable](#)) added members between 4.0 and 4.5 such that objects of these types cannot deserialize with .NET Framework 4.0 if they were serialized with .NET Framework 4.5. Note that if the

serialized data is both serialized and deserialized with the same .NET Framework version, no issue will occur.

Suggestion

[System.Runtime.Serialization.Formatters.Soap.SoapFormatter](#) serialization should be replaced with a serializer that is resilient to .NET Framework changes. Examples include [System.Text.Json](#) and [System.Runtime.Serialization.DataContractSerializer](#).

Warning

Binary serialization with `BinaryFormatter` can be dangerous. For more information, see the [BinaryFormatter security guide](#) and the [BinaryFormatter migration guide](#).

Warning

Do not confuse [DataContractSerializer](#) with [NetDataContractSerializer](#). [NetDataContractSerializer](#) is identified as a [dangerous serializer](#).

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [SoapFormatter.Serialize\(Stream, Object\)](#)
- [\[SoapFormatter.Serialize\(Stream, Object, Header\[\]\)\]\(https://learn.microsoft.com/en-us/dotnet/api/system.runtime.serialization.formatters.soap.soapformatter.serialize#system-runtime-serialization-formatters-soap-soapformatter-serialize\(system-io-stream-system-object-system-runtime-remoting-messaging-header\(\)\)\)](#)
- [SoapFormatter.Deserialize\(Stream\)](#)
- [SoapFormatter.Deserialize\(Stream, HeaderHandler\)](#)

XmlSerializer fails while serializing a type that hides an accessible member with an inaccessible one

Details

When serializing a derived type, the [System.Xml.Serialization.XmlSerializer](#) can fail if the type contains an inaccessible field or property that hides (via the 'new' keyword) a field or property of the same name that was previously accessible (public, for example) on the base type.

Suggestion

This problem can be solved by making the new, hiding member accessible to the [System.Xml.Serialization.XmlSerializer](#) (by marking it public, for example). Alternatively, the following config setting will revert to 4.0 [System.Xml.Serialization.XmlSerializer](#) behavior, which will fix the problem:

```
<system.xml.serialization>
<xmlSerializer useLegacySerializerGeneration="true" />
</system.xml.serialization>
```

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [XmlSerializer.Serialize\(Stream, Object\)](#)
- [XmlSerializer.Serialize\(TextWriter, Object\)](#)
- [XmlSerializer.Serialize\(Object, XmlSerializationWriter\)](#)
- [XmlSerializer.Serialize\(XmlWriter, Object\)](#)
- [XmlSerializer.Serialize\(Stream, Object, XmlSerializerNamespaces\)](#)
- [XmlSerializer.Serialize\(TextWriter, Object, XmlSerializerNamespaces\)](#)
- [XmlSerializer.Serialize\(XmlWriter, Object, XmlSerializerNamespaces\)](#)
- [XmlSerializer.Serialize\(XmlWriter, Object, XmlSerializerNamespaces, String\)](#)
- [XmlSerializer.Serialize\(XmlWriter, Object, XmlSerializerNamespaces, String, String\)](#)

Web Applications

Managed browser hosting controls from the .NET Framework 1.1 and 2.0 are blocked

Details

Hosting these controls is blocked in Internet Explorer.

Suggestion

Internet Explorer will fail to launch an application that uses managed browser hosting controls. The previous behavior can be restored by setting the `EnableIEHosting` value of the registry subkey `HKLM/SOFTWARE/MICROSOFT/.NETFramework` to `1` for x86 systems and for 32-bit processes on x64 systems, and by setting the `EnableIEHosting` value of the registry subkey `HKLM/SOFTWARE/Wow6432Node/Microsoft/.NETFramework` to `1` for 64-bit processes on x64 systems.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Windows Communication Foundation (WCF)

Error codes for `maxRequestLength` or `maxReceivedMessageSize` are different

Details

Messages in WCF web services hosted in Internet Information Services (IIS) or ASP.NET Development Server that exceed `maxRequestLength` (in ASP.NET) or `maxReceivedMessageSize` (in WCF) have different error code. The HTTP status code has changed from 400 (Bad Request) to 413 (Request Entity Too Large), and messages that exceed either the `maxRequestLength` or the `maxReceivedMessageSize` setting throw a [System.ServiceModel.ProtocolException](#) exception. This includes cases in which the transfer mode is Streamed.

Suggestion

This change facilitates debugging in cases where the message length exceeds the limits allowed by ASP.NET or WCF. You must modify any code that performs processing based on an HTTP 400 status code.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

System.ServiceModel.Web.WebServiceHost object no longer adds a default endpoint

Details

The [WebServiceHost](#) object no longer adds a default endpoint if an explicit endpoint has been added by application code.

Suggestion

If users will expect to be able to connect to a default endpoint and other explicit endpoints have been added to the [System.ServiceModel.Web.WebServiceHost](#), default endpoints should also be added explicitly (using [System.ServiceModel.ServiceHostBase.AddDefaultEndpoints\(\)](#)).

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [ServiceHost.AddServiceEndpoint\(Type, Binding, String\)](#)
- [ServiceHost.AddServiceEndpoint\(Type, Binding, Uri\)](#)
- [ServiceHost.AddServiceEndpoint\(Type, Binding, String, Uri\)](#)
- [ServiceHost.AddServiceEndpoint\(Type, Binding, Uri, Uri\)](#)
- [ServiceHost.AddServiceEndpoint\(Type, Binding, Uri, Uri\)](#)
- [ServiceHostBase.AddServiceEndpoint\(ServiceEndpoint\)](#)
- [ServiceHostBase.AddServiceEndpoint\(String, Binding, String\)](#)
- [ServiceHostBase.AddServiceEndpoint\(String, Binding, Uri\)](#)
- [ServiceHostBase.AddServiceEndpoint\(String, Binding, String, Uri\)](#)

- [ServiceHostBase.AddServiceEndpoint\(String, Binding, Uri, Uri\)](#)

The Replace method in OData URLs is disabled by default

Details

Beginning in the .NET Framework 4.5, the Replace method in OData URLs is disabled by default. When OData Replace is disabled (now by default), any user requests including replace functions (which are uncommon) will fail.

Suggestion

If the replace method is required (which is uncommon), it can be re-enabled through a config settings ([System.Data.Services.Configuration.DataServicesFeaturesSection.ReplaceFunction](#)). However, an enabled replace method can open security vulnerabilities and should only be used after careful review.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [System.Data.Services.DataService<T>](#)

Windows Forms

PreviewLostKeyboardFocus is called repeatedly if its handler shows a Windows Forms message box

Details

Beginning in the .NET Framework 4.5, calling [MessageBox.Show](#) from a [PreviewLostKeyboardFocus](#) handler will cause the handler to re-fire when the message box is closed, potentially resulting in an infinite loop of message boxes.

Suggestion

There are two options to work around this issue:

- It may be avoided by calling [MessageBox.Show](#) instead of [MessageBox.Show](#).
- It may be avoided by showing the message box from a [LostKeyboardFocus](#) event handler (as opposed to a [System.Windows.UIElement.PreviewLostKeyboardFocus](#) event handler).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [ContentElement.PreviewLostKeyboardFocus](#)
- [InputElement.PreviewLostKeyboardFocus](#)
- [UIElement.PreviewLostKeyboardFocus](#)
- [UIElement3D.PreviewLostKeyboardFocus](#)

WinForm's CheckForOverflowUnderflow property is now true for System.Drawing

Details

The CheckForOverflowUnderflow property for the System.Drawing.dll assembly is set to true.

Suggestion

Previously when overflows occurred, the result would be silently truncated. Now an [System.OverflowException](#) exception is thrown.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Windows Presentation Foundation (WPF)

Accessing a WPF DataGrid's selected items from a handler of the DataGrid's UnloadingRow event can cause a NullReferenceException

Details

Due to a bug in the .NET Framework 4.5, event handlers for [DataGrid](#) events involving the removal of a row can cause a [System.NullReferenceException](#) to be thrown if they access the [DataGrid's System.Windows.Controls.Primitives.Selector.SelectedItem](#) or [System.Windows.Controls.Primitives.MultiSelector.SelectedItems](#) properties.

Suggestion

This issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [DataGrid.UnloadingRow](#)
- [DataGrid.UnloadingRowDetails](#)

Calling DataGrid.CommitEdit from a CellEditEnding handler drops focus

Details

Calling [CommitEdit\(\)](#) from one of the [System.Windows.Controls.DataGrid's System.Windows.Controls.DataGrid.CellEditEnding](#) event handlers causes the [System.Windows.Controls.DataGrid](#) to lose focus.

Suggestion

This bug has been fixed in the .NET Framework 4.5.2, so it can be avoided by upgrading the .NET Framework. Alternatively, it can be avoided by explicitly re-selecting the [System.Windows.Controls.DataGrid](#) after calling [System.Windows.Controls.DataGrid.CommitEdit\(\)](#).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [DataGrid.CommitEdit\(\)](#)
- [DataGrid.CommitEdit\(DataGridEditingUnit, Boolean\)](#)

Calling Items.Refresh on a WPF ListBox, ListView, or DataGrid with items selected can cause duplicate items to appear in the element

Details

In the .NET Framework 4.5, calling [ListBox.Items.Refresh](#) from code while items are selected in a [System.Windows.Controls.ListBox](#) can cause the selected items to be duplicated in the list. A similar issue occurs with [System.Windows.Controls.ListView](#) and [System.Windows.Controls.DataGrid](#). This is fixed in the .NET Framework 4.6.

Suggestion

This issue may be worked around by programmatically unselecting items before [System.Windows.Data.CollectionView.Refresh\(\)](#) is called and then re-selecting them after the call is completed. Alternatively, this issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Value	Scope	Minor	Version	4.5	Type	Runtime
-------	-------	-------	---------	-----	------	---------

Affected APIs

- [CollectionView.Refresh\(\)](#)

FlowDocument may show an extra line of text

Details

In some cases, a [FlowDocument](#) element will display an extra line of text when running on the .NET Framework 4.5 compared to how it displayed when run on the .NET Framework 4.0. There are no known cases of the change causing any text to be displayed poorly or illegibly, but it could cause text to appear that previously was omitted from a [FlowDocument's](#) view.

Suggestion

In some cases, decreasing the display element's [PageHeight](#) property by one can restore the previous number of displayed lines.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [FlowDocument\(\)](#)
- [FlowDocument\(Block\)](#)
- [FlowDocumentReader\(\)](#)
- [FlowDocumentPageViewer\(\)](#)
- [DocumentPageView\(\)](#)

GlyphRun.ComputeInkBoundingBox() and FormattedText.Extent return different values beginning in .NET Framework 4.5

Details

Improvements were made to [ComputeInkBoundingBox\(\)](#) and [Extent](#) in the .NET Framework 4.5 to address issues where the boxes were too small for the contained glyphs in some cases in the .NET Framework 4.0. As a result of this, some bounding boxes will be larger beginning in the .NET Framework 4.5, resulting in subtle differences in UI layout.

Suggestion

Be aware that some glyph bounding box sizes have increased. These changes will usually improve presentation and hit box testing, but if the older (pre-.NET 4.5) behavior is desired, it can be opted into by adding the following entry to the app.config file:

```
<appsettings>
  <add key="IncludeAllInkInBoundingBox" value="false">
</appsettings>
```

Name	Value	Scope	Edge	Version	4.5	Type	Runtime	
------	-------	-------	------	---------	-----	------	---------	--

Affected APIs

- [GlyphRun.ComputeInkBoundingBox\(\)](#)
- [FormattedText.Extent](#)

Intermittently unable to scroll to bottom item in ItemsControls (like ListBox and DataGrid) when using custom DataTemplates

Details

In some instances, a bug in the .NET Framework 4.5 is causing ItemsControls (like [System.Windows.Controls.ListBox](#), [System.Windows.Controls.ComboBox](#), [System.Windows.Controls.DataGrid](#), etc.) to not scroll to their bottom item when using custom DataTemplates. If the scrolling is attempted a second time (after scrolling back up), it will work then.

Suggestion

This issue has been fixed in the .NET Framework 4.5.2 and may be addressed by upgrading to that version (or a later version) of the .NET Framework. Alternatively, users can still drag scroll bars to the final items in these collections, but may need to try twice to do so successfully.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Items.Clear does not remove duplicates from SelectedItems

Details

Suppose a Selector (with multiple selection enabled) has duplicates in its [System.Windows.Controls.Primitives.MultiSelector.SelectedItems](#) collection - the same item appears more than once. Removing those items from the data source (e.g. by calling `Items.Clear`) fails to remove them from [System.Windows.Controls.Primitives.MultiSelector.SelectedItems](#); only the first instance is removed. Furthermore, subsequent use of [System.Windows.Controls.Primitives.MultiSelector.SelectedItems](#) (e.g. `SelectedItems.Clear()`) can encounter problems such as [System.ArgumentException](#), because [System.Windows.Controls.Primitives.MultiSelector.SelectedItems](#) contains items that are no longer in the data source.

Suggestion

Upgrade if possible to .NET Framework 4.6.2.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [MultiSelector.SelectedItems](#)

ListBoxItem IsSelected binding issue with ObservableCollection<T>.Move

Details

Calling `Move(Int32, Int32)` or `MoveItem(Int32, Int32)` on a collection bound to a [System.Windows.Controls.ListBox](#) with items selected can lead to erratic behavior with future selection or unselection of [System.Windows.Controls.ListBox](#) items.

Suggestion

Calling [System.Collections.ObjectModel.Collection<T>.Remove\(T\)](#) and [System.Collections.ObjectModel.Collection<T>.Insert\(Int32, T\)](#) instead of `Move(Int32, Int32)` will work around this issue. Alternatively, this issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [ObservableCollection<T>.Move\(Int32, Int32\)](#)
- [ObservableCollection<T>.MoveItem\(Int32, Int32\)](#)

New enum values in WPF's PageRangeSelection

Details

Two new members ([System.Windows.Controls.PageRangeSelection.CurrentPage](#) and [System.Windows.Controls.PageRangeSelection.SelectedPages](#)) have been added to the [System.Windows.Controls.PageRangeSelection](#) enum.

Suggestion

In most cases, these changes won't impact user code. Code that depends on a particular number of elements existing in [GetNames\(Type\)](#) or [GetValues\(Type\)](#) calls on the [System.Windows.Controls.PageRangeSelection](#) type should be modified, though.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [System.Windows.Controls.PageRangeSelection](#)

PreviewLostKeyboardFocus is called repeatedly if its handler shows a Windows Forms message box

Details

Beginning in the .NET Framework 4.5, calling [MessageBox.Show](#) from a [PreviewLostKeyboardFocus](#) handler will cause the handler to re-fire when the message box is closed, potentially resulting in an infinite loop of message boxes.

Suggestion

There are two options to work around this issue:

- It may be avoided by calling [MessageBox.Show](#) instead of [MessageBox.Show](#).
- It may be avoided by showing the message box from a [LostKeyboardFocus](#) event handler (as opposed to a [System.Windows.UIElement.PreviewLostKeyboardFocus](#) event handler).

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [ContentElement.PreviewLostKeyboardFocus](#)
- [InputElement.PreviewLostKeyboardFocus](#)
- [UIElement.PreviewLostKeyboardFocus](#)
- [UIElement3D.PreviewLostKeyboardFocus](#)

Details

Right-clicking a selected [System.Windows.Controls.DataGrid](#) row header while multiple rows are selected results in the [System.Windows.Controls.DataGrid](#)'s selection changing to only that row.

Suggestion

This issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [DataGrid\(\)](#)

Scrolling a WPF TreeView or grouped ListBox in a VirtualizingStackPanel can cause the application to stop responding

Details

In the .NET Framework v4.5, scrolling a WPF [System.Windows.Controls.TreeView](#) in a virtualized stack panel can cause the application to stop responding if there are margins in the viewport (between the items in the [System.Windows.Controls.TreeView](#), for example, or on an ItemsPresenter element). Additionally, in some cases, different sized items in the view can cause instability even if there are no margins.

Suggestion

This bug can be avoided by upgrading to .NET Framework 4.5.1. Alternatively, margins can be removed from view collections (like [System.Windows.Controls.TreeViews](#)) within virtualized stack panels if all contained items are the same size.

Name	Value	Scope	Major	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [VirtualizingStackPanel.SetIsVirtualizing\(DependencyObject, Boolean\)](#)

WPF DataTemplate elements are now visible to UIA

Details

Previously, [System.Windows.DataTemplate](#) elements were invisible to UI Automation. Beginning in 4.5, UI Automation will detect these elements. This is useful in many cases, but can break tests that depend on UIA trees not containing [System.Windows.DataTemplate](#) elements.

Suggestion

UI Automation tests for this app may need updated to account for the UIA tree now including previously invisible [System.Windows.DataTemplate](#) elements. For example, tests that expect some elements to be next to each other may now need to expect previously invisible UIA elements in between. Or tests that rely on certain counts or indexes for UIA elements may need updated with new values.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [DataTemplate\(\)](#)
- [DataTemplate\(Object\)](#)

WPF DispatcherSynchronizationContext.CreateCopy now returns a new copy instead of the current instance

Details

In the .NET Framework 4, [CreateCopy\(\)](#) returned a reference to the current instance, primarily as a performance optimization. In the .NET Framework 4.5, it returns a new instance which makes it possible for the first time to conclude that equal references indicate the executing thread is in the correct synchronization context. It is unlikely that code that checks the identity of these references will be affected, but because of the change, code that calls [CreateCopy\(\)](#) should be tested as part of migration to the .NET Framework 4.5 or newer.

Suggestion

Be aware that [CreateCopy\(\)](#) will now return a new [System.Threading.SynchronizationContext](#) object. Previously, code that used equivalence of references generated this way was not actually checking whether it was in the proper context, but does when built against .NET Framework 4.5 or later. While unlikely to cause issues, exercising the affected code paths should be enough to determine if this poses any problem.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [DispatcherSynchronizationContext.CreateCopy\(\)](#)

WPF TextBox defaults to undo limit of 100

Details

In .NET Framework 4.5, the default undo limit for a WPF textbox is 100 (as opposed to being unlimited in .NET Framework 4.0)

Suggestion

If an undo limit of 100 is too low, the limit can be set explicitly with [UndoLimit](#)

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [System.Windows.Controls.TextBox](#)

WPF TextBox selected text appears a different color when the text box is inactive

Details

In .NET Framework 4.5, when a WPF text box control is inactive (it doesn't have focus), the selected text inside the box will appear a different color than when the control is active.

Suggestion

The previous (.NET Framework 4.0) behavior may be restored by setting the [AreInactiveSelectionHighlightBrushKeysSupported](#) property to `false`.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [System.Windows.Controls.TextBox](#)

WPF TreeViewItem must be used within a TreeView

Details

A change was introduced in 4.5 that restricts usage of [System.Windows.Controls.TreeViewItem](#) elements outside of a [System.Windows.Controls.TreeView](#). This manifests under the following conditions:

- [System.Windows.Controls.TreeViewItem](#)'s visual parent is not a panel. (A [System.Windows.Controls.TreeViewItem](#) generated for a [System.Windows.Controls.TreeView](#) will have a panel as its parent)
- The [System.Windows.Controls.TreeViewItem](#) is a descendant of a [System.Windows.Controls.VirtualizingStackPanel](#) acting as the "items host" for a list control (ListBox, DataGrid, ListView, etc.). Virtualization doesn't need to be enabled.
- The [System.Windows.Controls.VirtualizingStackPanel](#) is item-scrolling (`ScrollUnit="Item"`).
- Someone calls `VirtualizingStackPanel.MakeVisible(v)` to scroll an element `v` into view. This can be done explicitly, or implicitly in a number of ways; perhaps the most common way is simply clicking on `v` to give it the keyboard focus.
- The visual-parent chain from `v` to the [System.Windows.Controls.VirtualizingStackPanel](#) passes through the [System.Windows.Controls.TreeViewItem](#).

In other words, this is seen when a [System.Windows.Controls.TreeViewItem](#) is used outside of a [System.Windows.Controls.TreeView](#), and the user clicks on a descendant of the [System.Windows.Controls.TreeViewItem](#) to bring it into view. If the [System.Windows.Controls.TreeViewItem](#)

[eViewItem](#) has no focusable descendants, you'll never see this issue. An example of a situation where this is hit is when a [System.Windows.Controls.TreeViewItem](#) is the root of a [DataTemplate](#). When this issue is hit, there is an [InvalidCastException](#) that occurs within the WPF framework.

Suggestion

A hotfix will be made available for this.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Windows Workflow Foundation (WF)

System.Activities is now APTCA

Details

The assembly is marked with the [System.Security.AllowPartiallyTrustedCallersAttribute](#) attribute.

Suggestion

Derived classes cannot be marked with the [System.Security.SecurityCriticalAttribute](#). Previously, derived types had to be marked with the [System.Security.SecurityCriticalAttribute](#). However, this change should have no real impact.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

WF serializes Expressions.Literal<T> DateTimes differently now (breaks custom XAML parsers)

Details

The associated [ValueSerializer](#) object will convert a [System.DateTime](#) or [System.DateTimeOffset](#) object whose [Second](#) and [System.DateTime.Millisecond](#) components are non-zero and (for a [System.DateTime](#) value) whose [Kind](#) property is not [Unspecified](#) to property element syntax instead of a string. This change allows [System.DateTime](#) and [System.DateTimeOffset](#) values to be round-tripped. Custom XAML parsers that assume that input XAML is in the attribute syntax will not function correctly.

Suggestion

This change allows [System.DateTime](#) and [System.DateTimeOffset](#) values to be round-tripped. Custom XAML parsers that assume that input XAML is in the attribute syntax will

not function correctly.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

XML, XSLT

XmlSchemaException now sets line positions properly

Details

If the [SetLineInfo](#) value is passed to the Load method and a validation error occurs, the [LineNumber](#) and [LinePosition](#) properties now contain line information.

Suggestion

Exception-handling code that assumes [LineNumber](#) and [LinePosition](#) will not be set should be updated since these properties will now be set properly when SetLineInfo is used while loading XML.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [LoadOptions.SetLineInfo](#)

XmlTextReader DTD entity expansion is limited to 10,000,000 characters

Details

DTD entity expansion is now limited to 10,000,000 characters. Loading XML files without DTD entity expansion or with limited DTD entity expansion is unaffected. Files with DTD entities that expand to more than 10,000,000 characters fail to load, and now throw an exception.

Suggestion

If the limit of DTD entity expansion is too low 10,000,000, the value can be overridden with the [MaxCharactersFromEntities](#) property. An [System.Xml.XmlReaderSettings](#) with the proper [System.Xml.XmlReaderSettings.MaxCharactersFromEntities](#) value can be passed to `XmlReader.Create` that takes [System.Xml.XmlReaderSettings](#) (ie. [Create\(String, XmlReaderSettings\)](#))

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [System.Xml.XmlTextReader](#)

- [XmlTextReader\(\)](#)
- [XmlTextReader\(Stream\)](#)
- [XmlTextReader\(Stream, XmlNameTable\)](#)
- [XmlTextReader\(Stream, XmlNodeType, XmlParserContext\)](#)
- [XmlTextReader\(TextReader\)](#)
- [XmlTextReader\(TextReader, XmlNameTable\)](#)
- [XmlTextReader\(String\)](#)
- [XmlTextReader\(String, Stream\)](#)
- [XmlTextReader\(String, Stream, XmlNameTable\)](#)
- [XmlTextReader\(String, TextReader\)](#)
- [XmlTextReader\(String, TextReader, XmlNameTable\)](#)
- [XmlTextReader\(String, XmlNameTable\)](#)
- [XmlTextReader\(String, XmlNodeType, XmlParserContext\)](#)
- [XmlTextReader\(XmlNameTable\)](#)

XSLT forward compat now works

Details

In the .NET Framework 4, XSLT 1.0 forward compatibility had the following issues:

- Loading a style sheet failed if its version was set to 2.0 and the parser encountered an unrecognized XSLT 1.0 construct.
- The `xsl:sort` construct failed to sort data if the style sheet version was set to 1.1. In the .NET Framework 4.5, these issues have been fixed, and XSLT 1.0 forward compatibility mode works properly.

Suggestion

Most apps should be unaffected, however data will be sorted differently in some cases now that `xsl:sort` is respected. If `xsl:sort` is used in 1.1 style sheets, confirm that apps were not depending on the unsorted order of data. If apps rely on the 4.0 sorting behavior, remove `xsl:sort` from the style sheet.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [System.Xml.Xsl.XslCompiledTransform](#)

XSLT style sheet exception message changed

Details

In the .NET Framework 4.5, the text of the error message when an XSLT file is too complex is "The style sheet is too complex." In previous versions, the error message was "XSLT compile error." Application code that depends on the text of the error message will no

longer work. However, the exception types remain the same, so this change should have no real impact.

Suggestion

Update any app code depending on the exception message from this error condition to expect the new message, or (even better) update the code to depend only on the exception type ([System.Xml.Xsl.XsltException](#)), which has not changed.

Name	Value		Scope	Edge		Version	4.5		Type	Runtime	
------	-------	--	-------	------	--	---------	-----	--	------	---------	--

Affected APIs

- [XslCompiledTransform.Load\(String\)](#)
- [XslCompiledTransform.Load\(Type\)](#)
- [XslCompiledTransform.Load\(XmlReader\)](#)
- [XslCompiledTransform.Load\(IXPathNavigable\)](#)
- [\[XslCompiledTransform.Load\(MethodInfo, Byte\[\], Type\[\]\)\]](#)([https://learn.microsoft.com/en-us/dotnet/api/system.xml.xsl.xslcompiledtransform.load#system-xml-xsl-xslcompiledtransform-load\(system-reflection-methodinfo-system-byte\(\)-system-type\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.xml.xsl.xslcompiledtransform.load#system-xml-xsl-xslcompiledtransform-load(system-reflection-methodinfo-system-byte()-system-type())))
- [XslCompiledTransform.Load\(String, XsltSettings, XmlResolver\)](#)
- [XslCompiledTransform.Load\(XmlReader, XsltSettings, XmlResolver\)](#)
- [XslCompiledTransform.Load\(IXPathNavigable, XsltSettings, XmlResolver\)](#)

.NET Framework 4.5.1

ADO.NET

ADO.NET now attempts to automatically reconnect broken SQL connections

Details

Beginning in .NET Framework 4.5.1, .NET Framework will attempt to automatically reconnect broken SQL connections. Although this will typically make apps more reliable, there are edge cases in which an app needs to know that the connection was lost so that it can take some action upon reconnection.

Suggestion

If this feature is undesirable due to compatibility concerns, it can be disabled by setting the [System.Data.SqlClient.SqlConnectionStringBuilder.ConnectRetryCount](#) property of a connection string (or [System.Data.SqlClient.SqlConnectionStringBuilder](#)) to 0.

Name	Value		Scope	Edge		Version	4.5.1		Type	Runtime	
------	-------	--	-------	------	--	---------	-------	--	------	---------	--

Affected APIs

- [IDbConnection.ConnectionString](#)
- [SqlConnection.ConnectionString](#)
- [ConnectionStringSettings.ConnectionString](#)
- [DbConnection.ConnectionString](#)
- [DbConnectionStringBuilder.ConnectionString](#)
- [SqlConnectionStringBuilder\(\)](#)
- [SqlConnectionStringBuilder\(String\)\)](#)
- [DbConnectionStringBuilder\(\)](#)
- [DbConnectionStringBuilder\(Boolean\)\)](#)

Core

A ConcurrentDictionary serialized in .NET Framework 4.5 with NetDataContractSerializer cannot be deserialized by .NET Framework 4.5.1 or 4.5.2

Details

Due to internal changes to the type, [ConcurrentDictionary<TKey,TValue>](#) objects that are serialized with the .NET Framework 4.5 using the [System.Runtime.Serialization.NetDataContractSerializer](#) cannot be deserialized in the .NET Framework 4.5.1 or in the .NET Framework 4.5.2. Note that moving in the other direction (serializing with the .NET Framework 4.5.x and deserializing with the .NET Framework 4.5) works. Similarly, all 4.x cross-version serialization works with the .NET Framework 4.6. Serializing and deserializing with a single version of the .NET Framework is not affected.

Suggestion

If it is necessary to serialize and deserialize a [System.Collections.Concurrent.ConcurrentDictionary<TKey,TValue>](#) between the .NET Framework 4.5 and .NET Framework 4.5.1/4.5.2, a different serializer like the [System.Runtime.Serialization.DataContractSerializer](#) should be used instead of the [System.Runtime.Serialization.NetDataContractSerializer](#). Alternatively, because this issue is addressed in the .NET Framework 4.6, it may be solved by upgrading to that version of the .NET Framework.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime

Affected APIs

Not detectable via API analysis.

ConcurrentQueue<T>.TryPeek can return an erroneous null via its out parameter

Details

In some multi-threaded scenarios, [System.Collections.Concurrent.ConcurrentQueue<T>.TryPeek\(T\)](#) can return true, but populate the out parameter with a null value (instead of the correct, peeked value).

Suggestion

This issue is fixed in the .NET Framework 4.5.1. Upgrading to that Framework will solve the issue.

Name	Value	Scope	Major	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [ConcurrentQueue<T>.TryPeek\(T\)](#)

COR_PRF_GC_ROOT_HANDLES are not being enumerated by profilers

Details

In the .NET Framework v4.5.1, the profiling API `RootReferences2()` is incorrectly never returning `COR_PRF_GC_ROOT_HANDLE` (they are returned as `COR_PRF_GC_ROOT_OTHER` instead). This issue is fixed beginning in the .NET Framework 4.6.

Suggestion

This issue has been fixed in the .NET Framework 4.6 and may be addressed by upgrading to that version of the .NET Framework.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

Not detectable via API analysis.

Deserialization of objects across appdomains can fail

Details

In some cases, when an app uses two or more app domains with different application bases, trying to deserialize objects in the logical call context across app domains throws an exception.

Suggestion

See [Mitigation: Deserialization of Objects Across App Domains](#)

Name	Value	Scope	Edge	Version	4.5.1	Type	Runtime
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Affected APIs

Not detectable via API analysis.

EventListener truncates strings with embedded nulls

Details

[System.Diagnostics.Tracing.EventListener](#) truncates strings with embedded nulls. Null characters are not supported by the [System.Diagnostics.Tracing.EventSource](#) class. The change only affects apps that use [System.Diagnostics.Tracing.EventListener](#) to read [System.Diagnostics.Tracing.EventSource](#) data in process and that use null characters as delimiters.

Suggestion

[System.Diagnostics.Tracing.EventSource](#) data should be updated, if possible, to not use embedded null characters.

Name	Value	Scope	Edge	Version	4.5.1	Type	Runtime
------	-------	-------	------	---------	-------	------	---------

Affected APIs

- [EventListener\(\)](#)
- [EventListener.EnableEvents\(EventSource, EventLevel\)](#)
- [EventListener.EnableEvents\(EventSource, EventLevel, EventKeywords\)](#)
- [EventListener.EnableEvents\(EventSource, EventLevel, EventKeywords, IDictionary<String,String>\)\)](#)

EventSource.WriteEvent impls must pass WriteEvent the same parameters that it received (plus ID)

Details

The runtime now enforces the contract that specifies the following: A class derived from [System.Diagnostics.Tracing.EventSource](#) that defines an ETW event method must call the base class `EventSource.WriteEvent` method with the event ID followed by the same arguments that the ETW event method was passed.

Suggestion

An [System.IndexOutOfRangeException](#) exception is thrown if an [System.Diagnostics.Tracing.EventListener](#) reads [System.Diagnostics.Tracing.EventSource](#) data in process for an event source that violates this contract.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime
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Affected APIs

Not detectable via API analysis.

Marshal.SizeOf and Marshal.PtrToStructure overloads break dynamic code

Details

Beginning in the .NET Framework 4.5.1, dynamically binding to the methods [SizeOf<T>\(\)](#), [SizeOf<T>\(T\)](#), [PtrToStructure\(IntPtr, Object\)](#), [PtrToStructure\(IntPtr, Type\)](#), [PtrToStructure<T>\(IntPtr\)](#), or [PtrToStructure<T>\(IntPtr, T\)](#), (via Windows PowerShell, IronPython, or the C# dynamic keyword, for example) can result in `MethodInvocationExceptions` because new overloads of these methods have been added that may be ambiguous to the scripting engines.

Suggestion

Update scripts to clearly indicate which overload should be used. This can typically be done by explicitly casting the methods' type parameters as [Type](#). See [this link](#) for more detail and examples of how to workaround the issue.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime

Affected APIs

Not detectable via API analysis.

Some .NET APIs cause first chance (handled) `EntryPointNotFoundException`

Details

In the .NET Framework 4.5, a small number of .NET methods began throwing first chance [System.EntryPointNotFoundException](#)s. These exceptions were handled within the .NET Framework, but could break test automation that did not expect the first chance exceptions. These same APIs break some ApiVerifier scenarios when `HighVersionLie` is enabled.

Suggestion

This bug can be avoided by upgrading to .NET Framework 4.5.1. Alternatively, test automation can be updated to not break on first-chance [System.EntryPointNotFoundException](#) exceptions.

Value	Scope	Edge	Version	4.5	Type	Runtime

Affected APIs

- [Debug.Assert\(Boolean\)](#)
- [Debug.Assert\(Boolean, String\)](#)
- [Debug.Assert\(Boolean, String, String\)](#)
- [\[Debug.Assert\(Boolean, String, String, Object\[\]\)\]](#)([https://learn.microsoft.com/en-us/dotnet/api/system.diagnostics.debug.assert#system-diagnostics-debug-assert\(system-boolean-system-string-system-string-system-object\(\)\)](https://learn.microsoft.com/en-us/dotnet/api/system.diagnostics.debug.assert#system-diagnostics-debug-assert(system-boolean-system-string-system-string-system-object())))
- [XmlSerializer\(Type\)](#)

WinRT stream adapters no long call FlushAsync automatically on close

Details

In Windows Store apps, Windows Runtime stream adapters no longer call the FlushAsync method from the Dispose method.

Suggestion

This change should be transparent. Developers can restore the previous behavior by writing code like this:

```
using (var stream = GetWindowsRuntimeStream() as Stream)
{
    // do something
    await stream.FlushAsync();
}
```

Name	Value	Scope	Transparent	Version	4.5.1	Type	Runtime	
------	-------	-------	-------------	---------	-------	------	---------	--

Affected APIs

Not detectable via API analysis.

Data

ADO.NET now attempts to automatically reconnect broken SQL connections

Details

Beginning in .NET Framework 4.5.1, .NET Framework will attempt to automatically reconnect broken SQL connections. Although this will typically make apps more reliable, there are edge cases in which an app needs to know that the connection was lost so that it can take some action upon reconnection.

Suggestion

If this feature is undesirable due to compatibility concerns, it can be disabled by setting the [System.Data.SqlClient.SqlConnectionStringBuilder.ConnectRetryCount](#) property of a connection string (or [System.Data.SqlClient.SqlConnectionStringBuilder](#)) to 0.

Name	Value	Scope	Edge	Version	4.5.1	Type	Runtime	
------	-------	-------	------	---------	-------	------	---------	--

Affected APIs

- [IDbConnection.ConnectionString](#)
- [SqlConnection.ConnectionString](#)

- [ConnectionStringSettings.ConnectionString](#)
- [DbConnection.ConnectionString](#)
- [DbConnectionStringBuilder.ConnectionString](#)
- [SqlConnectionStringBuilder\(\)](#)
- [SqlConnectionStringBuilder\(String\)\)](#)
- [DbConnectionStringBuilder\(\)](#)
- [DbConnectionStringBuilder\(Boolean\)\)](#)

Serialization

NetDataContractSerializer fails to deserialize a ConcurrentDictionary serialized with a different .NET version

Details

By design, the [System.Runtime.Serialization.NetDataContractSerializer](#) can be used only if both the serializing and deserializing ends share the same CLR types. Therefore, it is not guaranteed that an object serialized with one version of the .NET Framework can be deserialized by a different version. [System.Collections.Concurrent.ConcurrentDictionary<TKey,TValue>](#) is a type that is known to not to deserialize correctly if serialized with the .NET Framework 4.5 or earlier and deserialized with the .NET Framework 4.5.1 or later.

Suggestion

There are a number of possible work-arounds for this issue:

- Upgrade the serializing computer to use the .NET Framework 4.5.1, as well.
- Use [System.Runtime.Serialization.DataContractSerializer](#) instead of [System.Runtime.Serialization.NetDataContractSerializer](#) as this does not expect the exact same CLR types at both serializing and deserializing ends.
- Use [System.Collections.Generic.Dictionary<TKey,TValue>](#) instead of [System.Collections.Concurrent.ConcurrentDictionary<TKey,TValue>](#) since it does not exhibit this particular 4.5->4.5.1 break.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

- [NetDataContractSerializer.Deserialize\(Stream\)\)](#)

Windows Communication Foundation (WCF)

MinFreeMemoryPercentageToActiveService is now respected

Details

This setting establishes the minimum memory that must be available on the server before a WCF service can be activated. It is designed to prevent [System.OutOfMemoryException](#)

exceptions. In the .NET Framework 4.5, this setting had no effect. In the .NET Framework 4.5.1, the setting is observed.

Suggestion

An exception occurs if the free memory available on the web server is less than the percentage defined by the configuration setting. Some WCF services that successfully started and ran in a constrained memory environment may now fail.

Name	Value	Scope	Minor	Version	4.5.1	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

Not detectable via API analysis.

Windows Presentation Foundation (WPF)

Scrolling a WPF TreeView or grouped ListBox in a VirtualizingStackPanel can cause the application to stop responding

Details

In the .NET Framework v4.5, scrolling a WPF [System.Windows.Controls.TreeView](#) in a virtualized stack panel can cause the application to stop responding if there are margins in the viewport (between the items in the [System.Windows.Controls.TreeView](#), for example, or on an ItemsPresenter element). Additionally, in some cases, different sized items in the view can cause instability even if there are no margins.

Suggestion

This bug can be avoided by upgrading to .NET Framework 4.5.1. Alternatively, margins can be removed from view collections (like [System.Windows.Controls.TreeViews](#)) within virtualized stack panels if all contained items are the same size.

Name	Value	Scope	Major	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [VirtualizingStackPanel.SetIsVirtualizing\(DependencyObject, Boolean\)](#)

.NET Framework 4.5.2

ASP.NET

ASP.NET MVC now escapes spaces in strings passed in via route parameters

Details

In order to conform to RFC 2396, spaces in route paths are now escaped when populating action parameters from a route. So, whereas `/controller/action/some data` would previously match the route `/controller/action/{data}` and provide `some data` as the data parameter, it will now provide `some%20data` instead.

Suggestion

Code should be updated to unescape string parameters from a route. If the original URI is needed, it can be accessed with the [RequestUri.OriginalString](#) API.

Name	Value	Scope	Minor	Version	4.5.2	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

- [RouteAttribute\(String\)](#)

No longer able to set EnableViewStateMac to false

Details

ASP.NET no longer allows developers to specify `<pages enableViewStateMac="false"/>` or `<@Page EnableViewStateMac="false" %>`. The view state message authentication code (MAC) is now enforced for all requests with embedded view state. Only apps that explicitly set the `EnableViewStateMac` property to `false` are affected.

Suggestion

`EnableViewStateMac` must be assumed to be true, and any resulting MAC errors must be resolved (as explained in [this guidance](#), which contains multiple resolutions depending on the specifics of what is causing MAC errors).

Name	Value	Scope	Major	Version	4.5.2	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

Not detectable via API analysis.

Profiling ASP.NET MVC4 apps can lead to Fatal Execution Engine Error

Details

Profilers using NGEN /Profile assemblies may crash profiled ASP.NET MVC4 applications on startup with a 'Fatal Execution Engine Exception'

Suggestion

This issue is fixed in the .NET Framework 4.5.2. Alternatively, the profiler may avoid this issue by specifying `COR_PR_DISABLE_ALL_NGEN_IMAGES` in its event mask.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

Data

SqlConnection.Open fails on Windows 7 with non-IFS Winsock BSP or LSP present

Details

[Open\(\)](#) and [OpenAsync\(CancellationToken\)](#) fail in the .NET Framework 4.5 if running on a Windows 7 machine with a non-IFS Winsock BSP or LSP are present on the computer. To determine whether a non-IFS BSP or LSP is installed, use the `netsh WinSock Show Catalog` command, and examine every `Winsock Catalog Provider Entry` item that is returned. If the Service Flags value has the `0x20000` bit set, the provider uses IFS handles and will work correctly. If the `0x20000` bit is clear (not set), it is a non-IFS BSP or LSP.

Suggestion

This bug has been fixed in the .NET Framework 4.5.2, so it can be avoided by upgrading the .NET Framework. Alternatively, it can be avoided by removing any installed non-IFS Winsock LSPs.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

- [SqlConnection.Open\(\)](#)
- [SqlConnection.OpenAsync\(CancellationToken\)](#)

Entity Framework

EF no longer throws for QueryViews with specific characteristics

Details

Entity Framework no longer throws a [System.StackOverflowException](#) exception when an app executes a query that involves a QueryView with a 0..1 navigation property that attempts to include the related entities as part of the query. For example, by calling `.Include(e => e.RelatedNavProp)`.

Suggestion

This change only affects code that uses QueryViews with 1-0..1 relationships when running queries that call `.Include`. It improves reliability and should be transparent to almost all apps. However, if it causes unexpected behavior, you can disable it by adding the following entry to the `<appSettings>` section of the app's configuration file:

```
<add key="EntityFramework_SimplifyUserSpecifiedViews" value="false" />
```

Name	Value	Scope	Edge	Version	4.5.2	Type	Runtime
------	-------	-------	------	---------	-------	------	---------

Affected APIs

Not detectable via API analysis.

Opt-in break to revert from different 4.5 SQL generation to simpler 4.0 SQL generation

Details

Queries that produce JOIN statements and contain a call to a limiting operation without first using `OrderBy` now produce simpler SQL. After upgrading to .NET Framework 4.5, these queries produced more complicated SQL than previous versions.

Suggestion

This feature is disabled by default. If Entity Framework generates extra JOIN statements that cause performance degradation, you can enable this feature by adding the following entry to the `<appSettings>` section of the application configuration (app.config) file:

```
<add key="EntityFramework_SimplifyLimitOperations" value="true" />
```

Name	Value	Scope	Transparent	Version	4.5.2	Type	Runtime
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Affected APIs

Not detectable via API analysis.

Windows Presentation Foundation (WPF)

Calling `DataGrid.CommitEdit` from a `CellEditEnding` handler drops focus

Details

Calling `CommitEdit()` from one of the `System.Windows.Controls.DataGrid`'s `System.Windows.Controls.DataGrid.CellEditEnding` event handlers causes the `System.Windows.Controls.DataGrid` to lose focus.

Suggestion

This bug has been fixed in the .NET Framework 4.5.2, so it can be avoided by upgrading the .NET Framework. Alternatively, it can be avoided by explicitly re-selecting the `System.Windows.Controls.DataGrid` after calling `System.Windows.Controls.DataGrid.CommitEdit()`.

Name	Value	Scope	Edge	Version	4.5	Type	Runtime
------	-------	-------	------	---------	-----	------	---------

Affected APIs

- [DataGrid.CommitEdit\(\)](#)
- [DataGrid.CommitEdit\(DataGridEditingUnit, Boolean\)](#)

Intermittently unable to scroll to bottom item in ItemsControls (like ListBox and DataGrid) when using custom DataTemplates

Details

In some instances, a bug in the .NET Framework 4.5 is causing ItemsControls (like [System.Windows.Controls.ListBox](#), [System.Windows.Controls.ComboBox](#), [System.Windows.Controls.DataGrid](#), etc.) to not scroll to their bottom item when using custom DataTemplates. If the scrolling is attempted a second time (after scrolling back up), it will work then.

Suggestion

This issue has been fixed in the .NET Framework 4.5.2 and may be addressed by upgrading to that version (or a later version) of the .NET Framework. Alternatively, users can still drag scroll bars to the final items in these collections, but may need to try twice to do so successfully.

Name	Value	Scope	Minor	Version	4.5	Type	Runtime
------	-------	-------	-------	---------	-----	------	---------

Affected APIs

Not detectable via API analysis.

WPF spawns a wisptis.exe process which can freeze the mouse

Details

An issue was introduced in 4.5.2 that causes `wisptis.exe` to be spawned that can freeze mouse input.

Suggestion

A fix for this issue is available in a servicing release of the .NET Framework 4.5.2 (hotfix rollup 3026376), or by upgrading to the .NET Framework 4.6

Name	Value	Scope	Major	Version	4.5.2	Type	Runtime
------	-------	-------	-------	---------	-------	------	---------

Affected APIs

Not detectable via API analysis.

XML

XML parsing changes

Name	Value	Scope	Minor	Version	4.5.2	Type	Runtime
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Details

For security reasons, the following changes were introduced into XML parsing APIs:

- [XmlReaderSettings.MaxCharactersFromEntities](#) is set to 10 million when [XmlReaderSettings](#) is initialized.
- [XmlReaderSettings.XmlResolver](#) is set to `null` by default.

Note

[XmlReaderSettings](#) is used by all XML parsers, so while this change helps the [XmlReader](#) case, it also affects other scenarios.

Suggestion

To revert to the previous behavior, you can set a value in the registry. Add a DWORD value named `EnableLegacyXmlSettings` to the

`HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\NETFramework\XML` registry key, and set its value to `1`. You can also add the registry value in the `HKEY_CURRENT_USER` hive instead.

Affected APIs

- [System.Xml.XmlReaderSettings.MaxCharactersFromEntities](#)
- [System.Xml.XmlReaderSettings.XmlResolver](#)

In addition, any XML API that depends on [XmlResolver](#), either directly or indirectly, is affected.